Glass Chain

Software Architecture

2026/04/27

Project Description

Glass Chain is a blockchain wallet tracker focused on wallet addresses moving large sums of money. The tracker allows users to keep tabs on large investors and how they move their currencies.

Project Deliverables

An MVP will require the following functionalities:

1. Secure account creation allows users to register and sign in.
2. Integration with at least one blockchain network to capture and store persistent transaction data for tracked addresses (eg, Ethereum Virtual Machine (EVM) https://github.com/ipsilon/evmone).
3. A filtering engine that allows users to identify specific wallets based on defined parameters, such as minimum total holdings and historical activity metrics (e.g., trade volume over the past year).
4. A real-time portfolio overview that integrates an external pricing API to convert cryptocurrency values into real-world fiat values.
5. A chronological trading timeline showing all transactions made by a specific wallet, including the historical real-world values of both currencies at the precise time of each trade.
6. An alerting system that triggers notifications when currency prices cross user-defined thresholds or volatility markers.

Quality Attributes

Below are the main quality attributes the application will focus on. In addition to these, implement strong privacy and security measures to protect external API credentials.

1. Availability: The system is evaluated for accessibility via a web browser at all times, with basic tests to ensure the interface remains responsive during periods of high blockchain data ingestion.
2. Scalability: The system should be evaluated on its ability to maintain query performance and update frequency as the number of tracked wallets and the volume of blockchain data increase.
3. Interoperability: The platform's interoperability is evaluated by its ability to interface with and ingest data from multiple blockchain networks. The system should demonstrate a unified data schema that normalises transaction metadata from different protocols (e.g., an EVM-based chain like Polygon and a non-EVM chain) into a consistent format.